

Pathogen Data Network Global Survey of Pathogen Data Stakeholders

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Note: None of the questions was mandatory.

1 Participant information

1.1 Email Address

Answer type: Open-Ended Response

1.2 First Name

Answer type: Open-Ended Response

1.3 Last Name

Answer type: Open-Ended Response

1.4 I have reviewed the information above regarding the processing of personal data and consent to my personal data being processed for the purposes stated above. If you disagree, please exit the survey at this time.

Answer type: checkbox

1.5 Does your work or study involve pathogen data relevant to human health?

Possible answers:

- Yes
- No

1.6 Which of the following best describes your primary role in working with pathogen data?

Possible answers:

- Analyst, Researcher, or Public healthcare professional
- Data Steward
- Policymaker
- Trainer or Educator
- Software or Computational Developer
- Other

2 Likert type question: What are the barriers you face when working with pathogen data?

Please consider the following definitions:

- Significant Barrier: A frequent or major issue that significantly limits your ability to make the best possible use of pathogen data.
- Not a Significant Barrier: An infrequent or minor issue that does not substantially impact your ability to work with pathogen data.
- Not Applicable: This barrier does not apply to your use of pathogen data.

2.1 Possible barriers

- Restricted Access to Data:(e.g., paywalls, restricted databases, or lengthy and complex approval processes that limit timely access to pathogen data).
- Unclear Data-Sharing Policies:(e.g., ambiguity around what data can be shared, when it can be shared, and with whom, creating uncertainty and hesitation in data-sharing practices).
- Data Security Challenges:(e.g., limited access to secure infrastructure or tools required for handling sensitive pathogen data safely and in compliance with regulations).
- Data Aggregation Challenges:(e.g., pathogen data that is siloed across systems, organizations, or formats, making it difficult to combine and share even when no explicit policy prohibits it).
- Insufficient Technical Resources: (e.g., lack of access to high-performance computing systems or software, unreliable internet connectivity, or limited server capacity for managing pathogen data).

- Shortage of Skilled Personnel: (e.g., an insufficient number of data scientists, bioinformaticians, or other experts either within the team or at organizational level).
- Barriers to Training, Education, and Capacity Building: (e.g., limited availability of training programs or workshops, higher-education degree programs, or workforce development programs).
- Limited Funding for Work: (e.g., insufficient budget for conducting experiments, acquiring computational tools, or accessing subscription journals and services).
- Legal Barriers or Uncertainty:(e.g., restrictions or constraints imposed by data protection laws such as GDPR or HIPAA, which limit access, sharing, or use of pathogen data, including ambiguity in data sharing and compliance).
- Potential for Data Misuse: (e.g., concerns about unauthorized use, misinterpretation of shared data, or issues related to data sovereignty and ownership).
- Intellectual Property Restrictions: (e.g., data ownership claims or proprietary rights preventing open access or collaboration on pathogen datasets).
- Informed Consent Limitations:(e.g., difficulty obtaining or verifying consent for the use of clinical or personal data in research, particularly in international datasets).
- Inadequate Recognition for Data Contributions:(e.g., lack of citation, acknowledgment, or credit for researchers or institutions that contribute pathogen data to studies or publications).
- Trust Issues Between Stakeholders: (e.g., reluctance to share data due to fears of unfair competition, exploitation, or lack of equitable benefit-sharing among contributors).
- Difficulty in Translating Data Insights into Policy: (e.g., challenges in ensuring that insights derived from pathogen data reach policymakers or government organizations capable of turning them into actionable policies).
- Lack of Interdisciplinary Collaboration: (e.g., limited communication or cooperation between disciplines such as epidemiology, microbiology, genetics, and policymaking, which hinders holistic decision-making and solutions).
- Social and Public Perceptions:(e.g., challenges in communicating the rationale for your work, including clearly explaining its benefits and potential risks)

3 Further multiple choice questions

3.1 Which of the proposed barriers is the single most significant barrier for you?

Possible answers:

- None of the above
- Restricted Access to Data
- Unclear Data-Sharing Policies
- Data Security Challenges

- Data Aggregation Challenges
- Lack of Standardized Data Formats
- Inadequate Data Quality
- Limited Contextual Metadata
- Lack of Established Best Practices for Data Analysis
- Insufficient Technical Resources
- Shortage of Skilled Personnel
- Barriers to Training and Capacity Building
- Limited Funding for Work
- Legal Barriers or Uncertainty
- Potential for Data Misuse
- Intellectual Property Restrictions
- Informed Consent Limitations
- Inadequate Recognition for Data Contributions
- Trust Issues Between Stakeholders
- Difficulty in Translating Data Insights into Policy
- Lack of Interdisciplinary Collaboration
- Social and Public Perceptions
- Other

3.2 Which of the following topics would you prioritize for training and professional development?

Possible answers:

- Use of Pathogens Portal
- Introduction to Programming Languages
- Bioinformatics for Infectious Disease Research
- FAIR Data Management
- Infectious Disease Data Management
- Infectious Disease Domain Knowledge

- Visualization and Communication of Results
- Updates on Pathogen Data Topics
- Introduction to International Processes and Forums
- Ethical, Legal, and Social Implications (ELSI) for Data
- Ontologies and Metadata Standards
- Data Management Plans
- Open Science
- Dealing with Legislation Related to Infectious Disease Data Management
- Building Capacity for Pathogen Data Science
- Using Real-World Case Studies in Education
- Building Scalable Bioinformatics Pipelines
- Machine Learning and AI for Infectious Disease Research
- Interoperability in Pathogen Data Systems
- Other

3.3 What format(s) of training (e.g, webinar, in-person workshop, written documentation) are you likely to engage with?

Possible answers:

- Webinar
- Virtual Short Courses
- In-person Short Courses
- Self-Paced Online Learning
- Hackathon
- Documentation
- Help Desk
- Other

3.4 In what country do you work?

Possible answers:

- Afghanistan
- Albania
- Algeria
- Andorra
- Angola
- Antigua and Barbuda
- Argentina
- Armenia
- Australia
- Austria
- Azerbaijan
- Bahamas
- Bahrain
- Bangladesh
- Barbados
- Belarus
- Belgium
- Belize
- Benin
- Bhutan
- Bolivia (Plurinational State of)
- Bosnia and Herzegovina
- Botswana
- Brazil
- Brunei Darussalam
- Bulgaria

- Burkina Faso
- Burundi
- Cabo Verde
- Cambodia
- Cameroon
- Canada
- Central African Republic
- Chad
- Chile
- China
- Colombia
- Comoros
- Congo
- Costa Rica
- Côte D'Ivoire
- Croatia
- Cuba
- Cyprus
- Czech Republic
- Democratic People's Republic of Korea
- Democratic Republic of the Congo
- Denmark
- Djibouti
- Dominica
- Dominican Republic
- Ecuador
- Egypt
- El Salvador
- Equatorial Guinea

- Eritrea
- Estonia
- Ethiopia
- Fiji
- Finland
- France
- Gabon
- Gambia
- Georgia
- Germany
- Ghana
- Greece
- Grenada
- Guatemala
- Guinea
- Guinea Bissau
- Guyana
- Haiti
- Holy See
- Honduras
- Hungary
- Iceland
- India
- Indonesia
- Iran (Islamic Republic of)
- Iraq
- Ireland
- Israel
- Italy

- Jamaica
- Japan
- Jordan
- Kazakhstan
- Kenya
- Kiribati
- Kuwait
- Kyrgyzstan
- Lao People's Democratic Republic
- Latvia
- Lebanon
- Lesotho
- Liberia
- Libya
- Liechtenstein
- Lithuania
- Luxembourg
- Madagascar
- Malawi
- Malaysia
- Maldives
- Mali
- Malta
- Marshall Islands
- Mauritania
- Mauritius
- Mexico
- Micronesia (Federated States of)
- Monaco

- Mongolia
- Montenegro
- Morocco
- Mozambique
- Myanmar
- Namibia
- Nauru
- Nepal
- Netherlands
- New Zealand
- Nicaragua
- Niger
- Nigeria
- Norway
- Oman
- Pakistan
- Palau
- Panama
- Papua New Guinea
- Paraguay
- Peru
- Philippines
- Poland
- Portugal
- Qatar
- Republic of Korea
- Republic of Moldova
- Romania
- Russian Federation

- Rwanda
- Saint Kitts and Nevis
- Saint Lucia
- Saint Vincent and the Grenadines
- Samoa
- San Marino
- Sao Tome and Principe
- Saudi Arabia
- Senegal
- Serbia
- Seychelles
- Sierra Leone
- Singapore
- Slovakia
- Slovenia
- Solomon Islands
- Somalia
- South Africa
- South Sudan
- Spain
- Sri Lanka
- State of Palestine
- Sudan
- Suriname
- Swaziland
- Sweden
- Switzerland
- Syrian Arab Republic
- Tajikistan

- Thailand
- The former Yugoslav Republic of Macedonia
- Timor-Leste
- Togo
- Tonga
- Trinidad and Tobago
- Tunisia
- Turkey
- Turkmenistan
- Tuvalu
- Uganda
- Ukraine
- United Arab Emirates
- United Kingdom of Great Britain and Northern Ireland
- United Republic of Tanzania
- United States of America
- Uruguay
- Uzbekistan
- Vanuatu
- Venezuela (Bolivarian Republic of)
- Vietnam
- Yemen
- Zambia
- Zimbabwe

3.5 How would you classify your career stage?

Possible answers:

- Trainee
- Early Career
- Mid-Career
- Late Career
- Other

3.6 How would you classify the institution you are professionally affiliated with?

Possible answers:

- Higher Education Institution
- Research Institute
- Government Agency
- Industry
- Non-Profit or NGO
- Healthcare Institution
- Other

4 Open questions

4.1 What is the name of your institution?

Answer type: Open-Ended Response

4.2 Is there any other comment you would like to make?

Answer type: Open-Ended Response

5 PDN resources

5.1 What Pathogen Data Network resources or tools would be most important for you or your peers?

Possible answers:

- Pathogens Portal (Central)
- Reference Software Implementation to Deploy a Pathogens Portal Node
- Seed Money to Support Deployment of a Pathogens Portal Node
- Community of Practice for Pathogens Portal Nodes
- Core Data Hubs to Host and Privately Share Data with Colleagues
- Pre-Computed Datasets (Analyzed by the Pathogen Analysis System)
- Advanced Analysis Pipelines for Metagenomics
- An LLM Chatbot Prototype for Querying Bioinformatics Results
- Validated Standard Analysis Pipelines for Pathogens of Interest
- Metadata Standards
- Policy Briefs on Pathogen Data Sharing
- Open Community Forum
- Technical Capacity Framework (Maturity Model) to Assess Pathogen Data Platforms
- Webinars/Training on Emerging Laws, Regulations, and Policies
- In-Person Workshops for Integrating Pathogen Data into the Undergraduate Curriculum
- Experts
- Other